

01 — Executive Summary

SEION-KGR v26 MAX — Gate 12 Closeout Campaign

Campaign ID: `gate12-closeout-2026-08-01`

Canonical commit: `6af3c35271ae2ffab41ecba2aad098d1988fdc0c`

Branch: `campaign/gate12-closeout`

Date: `2026-08-01`

Decision-oriented summary. Read 09 before acting on anything here.

1 What was built

Starting from an implementation-complete, smoke-verified system (canonical commit `6af3c35271ae2ffab41ecba2aad098d1988fdc0c`), this campaign closed five engineering gaps the canonical commit explicitly left open: (1) a real GitHub Actions CI workflow, executed and green; (2) a genuinely learned sparse path selector, replacing the previous fixed-random budget; (3) a controlled E_8 structural-kernel residual branch with four matched controls, using the real E_8 Lie-algebra structure-constant tensor; (4) a state-to-score-to-ranking certification chain that correctly and honestly refuses to certify anything it cannot check; (5) a reproducibility audit that caught and fixed two real bugs (a configuration-identity hash that wrongly blocked ordinary training-extension resumes; a projector orthonormality bug introduced by an unnecessary sign-correction). Test count grew from 66 to 120, all passing; three real bugs were caught by tests written for other purposes and fixed during this session, not deferred.

2 What improved

Engineering correctness and auditability improved substantially and measurably (CI is green; reproducibility identity is enforced and tested; certification honestly reports near-zero coverage rather than an inflated one). The base-expert-only architecture (TuckER, selected via a real 1-epoch screening pass) reproduces expected qualitative KGE behavior on both FB15K-237 (test MRR 0.278/0.277 across 2 seeds) and WN18RR (0.441/0.424).

3 What did not improve, or was not tested

No claim is made that any of SEION-KGR’s experimental branches (path reasoner, seionic branch, projection, rank controller, E_8 residual) improve link-prediction quality. The one architectural comparison executed this campaign (base expert alone vs. base + path + seionic) is confounded by a large, logged training-data-size difference and is not informative about the architectural question it was meant to probe (§04, §09). No E_8 -vs-controls comparison was run on benchmark data. No confirmatory statistical campaign (5 seeds, full ablation ladder) was executed — this was declared as out of scope for a single session *before* the campaign began (`preregistration.md` §0), not discovered as a shortfall afterward.

4 Strongest valid result

The engineering closure itself: a real, CI-verified, 120-test, multi-branch KGE system with a working learned selector, a working E_8 /control residual branch, and a certification chain that is honest about its own limits, verified end-to-end on real benchmark data and real GPU hardware. This is a valid, load-bearing result. The benchmark numbers in §04 are not.

5 Readiness classification

Dimension	Status
Engineering / reproducibility	Closed — CI green, 120/120 tests pass, real GPU parity verified
Screening-tier benchmark evidence	Partial — 2 datasets, 2 ladder points, confounded comparison
Confirmatory statistical evidence	Not started
Causal ablation (full 13-point ladder)	Not started (2/13 points at screening scale only)
E_8 causal attribution	Not started
General-case certification	Open (narrow zero-compression case only)
Publication package	This document and its 11 companions

6 Next decision

Before any benchmark or ablation claim can be made: (a) resolve the path-reasoner scaling problem (§09) so A3-class configurations can run at full dataset scale within a tractable compute budget; (b) re-run the A0/A3 comparison on matched data scale and epoch budget; (c) only then proceed to the full preregistered screening/confirmatory protocol. Estimated compute for that full protocol, extrapolating from this campaign’s measurements, is multi-day, not multi-hour — a genuinely separate resourcing decision from this campaign.